

HARSH JHA

Node.js Backend Developer • Express • MongoDB • Redis • AWS

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PROFESSIONAL SUMMARY

Backend developer specializing in Node.js and Express, with hands-on experience building production-grade, transaction-safe financial systems and large-scale web automation pipelines. Skilled in designing concurrency-safe data models, RESTful APIs, and secure authentication flows across MongoDB, Redis, and PostgreSQL. Delivered automation covering 100+ U.S. county systems and a fintech ledger backend with zero-overdraft guarantees under simultaneous load.

EXPERIENCE

Backend & Automation Engineer Intern | Invicto Labs

Aug 2025 – Dec 2025

Bangalore, India

- Automated data retrieval across 100+ U.S. county websites using **Node.js** and **Puppeteer** to extract property tax and mortgage records at scale.
- Automated verification of **tax dues and payment history** via unique property identifiers, accelerating mortgage processing turnaround.
- Built **Express.js API endpoints** to trigger county-specific automation jobs and feed retrieved data into internal mortgage-verification workflows.
- Designed end-to-end **Puppeteer automation workflows** for multi-step county portal verification, eliminating manual navigation and cutting processing time.

Backend Developer Intern | TRIOT Solutions

Jan 2025 – Jul 2025

Remote, India

- Enhanced **Spring Boot** backend services by implementing **REST APIs** and integrating **PostgreSQL**, ensuring seamless data flow to a **React** and **Tailwind CSS** frontend.
- Designed and implemented **ride-booking workflows**, enabling owners to manage vehicle availability, booking requests, and trip assignments.
- Developed secure **BCrypt-based authentication** and role-based onboarding workflows, collaborating with frontend teams on responsive **Tailwind CSS** components.
- Collaborated in an **Agile** environment to integrate **Spring Boot** REST APIs with the **React** frontend, delivering end-to-end ride-booking workflows.

PROJECTS

Nivora Ledger — Production-Grade Fintech Backend

Node.js • Express • MongoDB • Redis • JWT • Zod • Jest

- Built a **double-entry ledger backend** with an **immutable, append-only ledger**, deriving account balances from ledger entries rather than mutable fields to eliminate balance drift by construction.
- Implemented **atomic fund transfers** using MongoDB **multi-document transactions** and **optimistic concurrency locking** (versioned accounts), preventing double-spend under simultaneous transfer attempts; enforced mandatory **idempotency keys** to block duplicate processing on retries.
- Wrote a **29-test Jest/Supertest suite** against a live single-node MongoDB replica set, including concurrency tests proving **zero overdraft** across simultaneous transfer attempts on the same account.
- Added **environment-configurable rate limiting** (per-email on auth, per-user on transfers) and a **paginated transaction history endpoint** with account-level access isolation.

TECHNICAL SKILLS

- Languages & Backend:** Node.js, Express.js, TypeScript, JavaScript, Java, Spring Boot, REST APIs, Jest, Zod, JWT
- Frontend:** React.js, Tailwind CSS
- Databases:** MongoDB, Redis, PostgreSQL, MySQL
- DevOps & Tools:** Docker, AWS (EC2, S3), GitHub Actions, Linux, Git
- Concepts:** Distributed Systems, Concurrency, Idempotency, Double-Entry Ledger, Rate Limiting

EDUCATION

Jain University — MCA

Jul 2024 – Present

Bangalore, India

Girijananda Chowdhury University — BCA

Aug 2020 – Jul 2023

Guwahati, India